Fixed-difference Kummer collector

KUMMER-DIFFERENTIAL-COLLECTOR-20261022

Completed targets per arm: 1260.

Uncached: 1260 source-verified, 230,430,039 quotient steps.

Uncached: 109,480,302 signed splits, 739 Recover failures.

Uncached cold ratio: 8.9597–55.9354; wins/losses: 0/36 cells.

Cached: 1260 source-verified, 230,430,039 quotient steps.

Cached: 109,480,302 signed splits, 739 Recover failures.

Cached cold ratio: 8.6176–54.0936; wins/losses: 0/36 cells.

Cold setup: 21 unique model builds, charged in all 1260 targets.

Source transfers and source verifications: 1260 each.

Every candidate and rho answer was independently checked as k·P=Q

on the original elliptic curve over Fp^2.

Rho is a same-source archived control from a separate process.

These ratios use archived rho counts; see KUMMER_PAIRED.pdf for matched counts.

Model setup includes root search, theta search and torsion matrices.

Each cold target includes source transfer, initialization, the

quotient walk, signed Recover, graph elimination and verification.

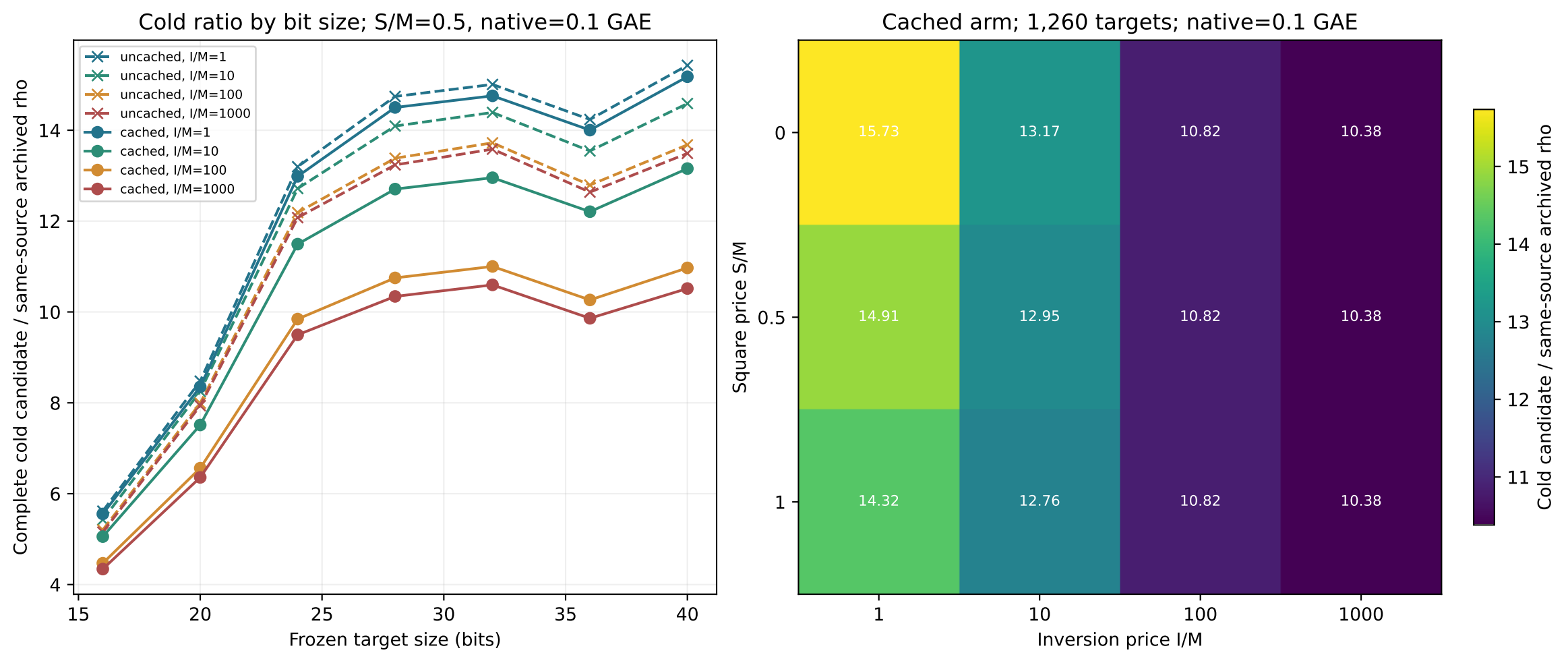
Raw: results/<arm>/collector-*.jsonl, cold-*-setup.jsonl,

results/target-price.csv and results/price-grid.csv.

All 36 cells use S/M in {0,0.5,1}, I/M in {1,10,100,1000},

and native cost in {0,0.1,1} GAE per declared logical operation.

Ratios describe this measured implementation and frozen cohort.


Cold ratio by bit size; S/M=0.5, native=0.1 GAE
uncached, I/M=1
uncached, I/M=10
uncached, I/M=100
uncached, I/M=1000
cached, I/M=1
cached, I/M=10
cached, I/M=100
cached, I/M=1000
Complete cold candidate / same-source archived rho
Frozen target size (bits)
Cached arm; 1,260 targets; native=0.1 GAE
Square price S/M
Inversion price I/M
15.73
13.17
10.82
10.38
14.91
12.95
10.82
10.38
14.32
12.76
10.82
10.38
Cold candidate / same-source archived rho

uncached: native=0 GAE/logical

S/M	I/M	candidate GAE	rho GAE	ratio	wins/losses
0.0	1	9,798,151,036.8	175,168,978.8	55.935	0/1260
0.0	10	8,378,420,339.1	174,640,420.7	47.975	0/1260
0.0	100	7,075,915,111.8	174,155,505.1	40.630	0/1260
0.0	1000	6,830,646,535.8	174,064,192.9	39.242	0/1260
0.5	1	8,841,142,700.8	175,041,121.6	50.509	0/1260
0.5	10	7,966,770,880.2	174,617,698.3	45.624	0/1260
0.5	100	7,021,504,047.1	174,159,943.5	40.316	0/1260
0.5	1000	6,825,157,820.6	174,064,860.9	39.210	0/1260
1.0	1	8,157,565,318.0	174,949,795.0	46.628	0/1260
1.0	10	7,626,712,631.6	174,598,927.7	43.681	0/1260
1.0	100	6,969,019,037.8	174,164,224.7	40.014	0/1260
1.0	1000	6,819,690,778.5	174,065,526.2	39.179	0/1260

uncached: native=0.1 GAE/logical

S/M	I/M	candidate GAE	rho GAE	ratio	wins/losses
0.0	1	17,462,948,021.6	1,089,353,650.8	16.031	0/1260
0.0	10	16,043,217,323.9	1,088,825,092.7	14.734	0/1260
0.0	100	14,740,712,096.6	1,088,340,177.1	13.544	0/1260
0.0	1000	14,495,443,520.6	1,088,248,864.9	13.320	0/1260
0.5	1	16,505,939,685.6	1,089,225,793.6	15.154	0/1260
0.5	10	15,631,567,865.0	1,088,802,370.3	14.357	0/1260
0.5	100	14,686,301,031.9	1,088,344,615.5	13.494	0/1260
0.5	1000	14,489,954,805.4	1,088,249,532.9	13.315	0/1260
1.0	1	15,822,362,302.8	1,089,134,467.0	14.527	0/1260
1.0	10	15,291,509,616.4	1,088,783,599.7	14.045	0/1260
1.0	100	14,633,816,022.6	1,088,348,896.7	13.446	0/1260
1.0	1000	14,484,487,763.3	1,088,250,198.2	13.310	0/1260

uncached: native=1 GAE/logical

S/M	I/M	candidate GAE	rho GAE	ratio	wins/losses
0.0	1	86,446,120,884.8	9,317,015,698.8	9.278	0/1260
0.0	10	85,026,390,187.1	9,316,487,140.7	9.126	0/1260
0.0	100	83,723,884,959.8	9,316,002,225.1	8.987	0/1260
0.0	1000	83,478,616,383.8	9,315,910,912.9	8.961	0/1260
0.5	1	85,489,112,548.8	9,316,887,841.6	9.176	0/1260
0.5	10	84,614,740,728.2	9,316,464,418.3	9.082	0/1260
0.5	100	83,669,473,895.1	9,316,006,663.5	8.981	0/1260
0.5	1000	83,473,127,668.6	9,315,911,580.9	8.960	0/1260
1.0	1	84,805,535,166.0	9,316,796,515.0	9.102	0/1260
1.0	10	84,274,682,479.6	9,316,445,647.7	9.046	0/1260
1.0	100	83,616,988,885.8	9,316,010,944.7	8.976	0/1260
1.0	1000	83,467,660,626.5	9,315,912,246.2	8.960	0/1260

cached: native=0 GAE/logical

S/M	I/M	candidate GAE	rho GAE	ratio	wins/losses
0.0	1	9,475,528,396.4	175,168,978.8	54.094	0/1260
0.0	10	6,680,406,442.2	174,640,420.7	38.252	0/1260
0.0	100	4,116,074,374.1	174,155,505.1	23.634	0/1260
0.0	1000	3,633,197,176.0	174,064,192.9	20.873	0/1260
0.5	1	8,572,290,500.5	175,041,121.6	48.973	0/1260
0.5	10	6,430,472,592.6	174,617,698.3	36.826	0/1260
0.5	100	4,114,993,773.2	174,159,943.5	23.628	0/1260
0.5	1000	3,634,033,781.0	174,064,860.9	20.877	0/1260
1.0	1	7,927,120,574.9	174,949,795.0	45.311	0/1260
1.0	10	6,224,005,499.4	174,598,927.7	35.647	0/1260
1.0	100	4,113,951,423.6	174,164,224.7	23.621	0/1260
1.0	1000	3,634,867,082.6	174,065,526.2	20.882	0/1260

cached: native=0.1 GAE/logical

S/M	I/M	candidate GAE	rho GAE	ratio	wins/losses
0.0	1	17,140,325,381.2	1,089,353,650.8	15.734	0/1260
0.0	10	14,345,203,427.0	1,088,825,092.7	13.175	0/1260
0.0	100	11,780,871,358.9	1,088,340,177.1	10.825	0/1260
0.0	1000	11,297,994,160.8	1,088,248,864.9	10.382	0/1260
0.5	1	16,237,087,485.3	1,089,225,793.6	14.907	0/1260
0.5	10	14,095,269,577.4	1,088,802,370.3	12.946	0/1260
0.5	100	11,779,790,758.0	1,088,344,615.5	10.824	0/1260
0.5	1000	11,298,830,765.8	1,088,249,532.9	10.383	0/1260
1.0	1	15,591,917,559.7	1,089,134,467.0	14.316	0/1260
1.0	10	13,888,802,484.2	1,088,783,599.7	12.756	0/1260
1.0	100	11,778,748,408.4	1,088,348,896.7	10.823	0/1260
1.0	1000	11,299,664,067.4	1,088,250,198.2	10.383	0/1260

cached: native=1 GAE/logical

S/M	I/M	candidate GAE	rho GAE	ratio	wins/losses
0.0	1	86,123,498,244.4	9,317,015,698.8	9.244	0/1260
0.0	10	83,328,376,290.2	9,316,487,140.7	8.944	0/1260
0.0	100	80,764,044,222.1	9,316,002,225.1	8.669	0/1260
0.0	1000	80,281,167,024.0	9,315,910,912.9	8.618	0/1260
0.5	1	85,220,260,348.5	9,316,887,841.6	9.147	0/1260
0.5	10	83,078,442,440.6	9,316,464,418.3	8.917	0/1260
0.5	100	80,762,963,621.2	9,316,006,663.5	8.669	0/1260
0.5	1000	80,282,003,629.0	9,315,911,580.9	8.618	0/1260
1.0	1	84,575,090,422.9	9,316,796,515.0	9.078	0/1260
1.0	10	82,871,975,347.4	9,316,445,647.7	8.895	0/1260
1.0	100	80,761,921,271.6	9,316,010,944.7	8.669	0/1260
1.0	1000	80,282,836,930.6	9,315,912,246.2	8.618	0/1260